

Paediatric Endoscopy, Part One: Indications and Preparation.

Picture a Monday endoscopy list in a children's hospital. The first child is nine years old and has had pain around the belly button for six months. She is growing well, her blood tests are normal, and her stool calprotectin is low. The second is a six-year-old boy whose coeliac antibodies came back fourteen times above the upper limit of normal. The third is a fourteen-year-old with bloody diarrhoea and weight loss. The fourth is a seven-kilogram infant who vomited blood last night. Each of these children has been booked for an endoscopy. By the end of this chapter you should be able to say which of them actually needs one, how each should be anaesthetised and fasted, how the colon should be prepared, and which instrument to pick up.

The clinical question is simple. Which child needs an endoscopy, and how do we get that child ready? The main idea is that in children an endoscopy almost always comes with a general anaesthetic, a period of fasting and, for the colon, a day or more of laxatives. Each of these carries a cost and a small risk, so each must be earned by a real indication. Once the indication is clear, the details of preparation decide whether the procedure gives a useful answer the first time.

We will work through five steps. First, why endoscopy in children is mainly a way of obtaining tissue. Second, the indications and, just as importantly, the situations in which the guidelines say the scope adds nothing. Third, anaesthesia, sedation, the airway and fasting. Fourth, bowel preparation. Fifth, the choice of instrument by weight. Then we compare the North American and European documents and finish with what to do differently on Monday.

A note on sources. The main document is the 2017 guideline on paediatric gastrointestinal endoscopy from the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, called ESPGHAN, and the European Society of Gastrointestinal Endoscopy, called ESGE, led by Thomson and Tringali. We also use the 2022 standards of the Paediatric Endoscopy Quality Improvement Network, known as PEnQuIN, the 2014 clinical report on bowel preparation from the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, called NASPGHAN, and several guidelines that tell us when endoscopy can be avoided: the 2020 ESPGHAN coeliac guideline, the 2018 joint reflux guideline, and the 2023 joint guideline on *Helicobacter pylori*.

Part one. The mucosa hides its diagnosis.

In adults, endoscopy is often about what the eye can see: a bleeding ulcer, a polyp, a tumour. In children, it is mostly about what the microscope can see. A careful study from Colorado, published by Sheiko, Kramer and colleagues in 2015, looked at the first diagnostic upper endoscopy in one thousand children. The endoscopist saw something abnormal in about thirty-five percent of them. The pathologist found something abnormal in about forty percent. But the two did not overlap well; their agreement was only fair, with a kappa statistic of zero point three two. If biopsies had been taken only from mucosa that looked abnormal, almost half of the histological disease would

have been missed. In children with eosinophilic oesophagitis, the mucosa looked normal in about thirty percent, and in children with coeliac disease it looked normal in about forty-three percent.

This is why the ESPGHAN and ESGE guideline asks for systematic biopsies. For suspected eosinophilic oesophagitis, take at least three biopsies, from the proximal, middle and distal oesophagus. For suspected coeliac disease, take at least four from the distal duodenum and at least one from the duodenal bulb. The practical message is that if the indication justifies putting a child under anaesthesia, it also justifies biopsies from mucosa that looks normal.

Part two. When to scope, and when not to.

The guideline lists diagnostic indications for upper endoscopy and for ileocolonoscopy. Its recommendation is formally weak and based on low-quality evidence, which is typical of paediatric endoscopy, but the lists are sensible. For upper endoscopy, the indications include weight loss or poor growth, unexplained anaemia, abdominal pain that suggests organic disease, difficulty or pain on swallowing, unexplained recurrent vomiting, vomiting of blood, suspected eosinophilic oesophagitis, coeliac disease or peptic disease, caustic ingestion, foreign bodies, upper gastrointestinal bleeding, graft-versus-host disease, and chronic reflux when another disease must be excluded or Barrett oesophagus must be watched. For ileocolonoscopy, the main indication is suspected inflammatory bowel disease. Others are unexplained anaemia or chronic diarrhoea, perianal lesions, rectal bleeding, unexplained poor growth, graft-versus-host disease, intestinal transplantation, polyposis syndromes and a stricture seen on imaging. What all of these share is an organic question that a biopsy, or a therapeutic act, can answer.

The guideline is just as clear about when not to scope. It does not suggest upper endoscopy for uncomplicated gastro-oesophageal reflux or for functional gastrointestinal disorders. It suggests against ileocolonoscopy in toxic megacolon, within twenty-eight days of a colonic perforation, within seven days of an intestinal resection, and in functional disorders. Constipation is listed as a non-indication.

Several disease-specific guidelines take children away from the endoscopy list. The first is reflux. The 2018 joint ESPGHAN and NASPGHAN reflux guideline, led by Rosen, suggests not using endoscopy to diagnose reflux disease in infants and children. Endoscopy is reserved for complications, for suspected mucosal disease such as eosinophilic oesophagitis, or before escalating therapy. Barium studies should not be used to diagnose reflux at all.

The second is coeliac disease. The 2020 ESPGHAN guideline, led by Husby, allows a diagnosis without biopsy when the tissue transglutaminase IgA antibody, often called TGA or tTG, is at least ten times the upper limit of normal, and when endomysial antibody, called EMA, is positive in a second blood sample. HLA typing and symptoms are no longer required. The evidence comes from the ProCeDE study, published by Werkstetter and colleagues in 2017. It enrolled seven hundred and seven children in thirty-three centres across twenty-one countries. Three hundred and ninety-nine of them met the criteria of a TGA at least ten times normal, a positive EMA, and any symptom. In that group the positive predictive value for coeliac disease was ninety-nine point

seven five percent, and in children with malabsorption symptoms it was one hundred percent. Adding HLA typing made no difference. More than half of children with coeliac disease could avoid an endoscopy. Below ten times normal, a biopsy is still required, with the four plus one sampling pattern.

The third is *Helicobacter pylori*. The 2023 joint ESPGHAN and NASPGHAN guideline, published by Homan and colleagues in 2024, says that there is no evidence to support a test-and-treat strategy in children. It recommends against testing for *Helicobacter* in functional abdominal pain, in the initial work-up of iron deficiency anaemia, and in chronic immune thrombocytopenia. Here the rule runs the other way from reflux: if you do decide to look for *Helicobacter*, the diagnosis should be based on endoscopy, with at least six biopsies, three from the antrum and three from the body, so that the organism can be cultured or tested by polymerase chain reaction, PCR, for clarithromycin resistance before treatment is chosen.

The fourth tool is faecal calprotectin, a protein released by neutrophils into the stool. In 2010, van Rhee and colleagues published a meta-analysis in the British Medical Journal. In seven paediatric studies with three hundred and seventy-one children, calprotectin had a sensitivity of ninety-two percent and a specificity of seventy-six percent for inflammatory bowel disease. In adults the specificity was much higher, ninety-six percent. On a bar chart, the sensitivity bars are almost equal in children and adults, while the specificity bar for children is clearly shorter. The authors calculated that using calprotectin to decide who needs a scope would mean that sixty-five out of every one hundred children with suspected inflammatory bowel disease would undergo endoscopy instead of all one hundred, at the cost of a delayed diagnosis in about eight percent. A later paediatric meta-analysis by Degraeuwe and colleagues in 2015 found a sensitivity of ninety-seven percent and a specificity of seventy percent. Specificity is lower in children because infections, anti-inflammatory drugs, polyps and very young age all raise calprotectin.

Finally, what about the most common referral of all, abdominal pain? The data are mixed. In one series by Thakkar and colleagues, about thirty-eight percent of upper endoscopies for abdominal pain were diagnostic, with reflux oesophagitis the most common finding at about twenty-one percent, and in a smaller study the endoscopy changed management in about two thirds of children. But finding mild oesophagitis or gastritis does not mean it explains the pain. In a comparison summarised by Llanos-Chea and Saps, symptoms persisted at follow-up in sixty-one percent of children who had an endoscopy and sixty-four percent of those who did not. When children are selected by agreed criteria, the yield is higher: in a Malaysian study by Lee and colleagues, almost all procedures met the criteria and about three quarters showed positive findings. So in a child whose pain sounds functional, with normal growth, normal blood tests, normal calprotectin and no alarm features, the scope rarely changes the outcome.

Part three. Anaesthesia, the airway and fasting.

The ESPGHAN and ESGE guideline suggests that upper endoscopy and ileocolonoscopy in children be done under general anaesthesia or, only if general anaesthesia is not available, under

deep sedation in a carefully monitored environment. For endoscopic retrograde cholangiopancreatography, called ERCP, general anaesthesia is suggested, although deep or conscious sedation can be considered in teenagers between twelve and seventeen years of age. For bleeding, especially from varices, general anaesthesia is recommended.

The reason is safety and success. The large North American database called PEDS-CORI was analysed by Thakkar and colleagues in 2007. Among ten thousand two hundred and thirty-six upper endoscopies in children, immediate complications occurred in about two percent. They occurred in one point two percent of procedures under general anaesthesia and in three point seven percent under intravenous sedation, about three times as often. Four in five complications were cardiopulmonary, mostly brief drops in oxygen saturation, and none was fatal. Practice has shifted strongly towards anaesthesia since then. As a historical contrast, a survey of French-speaking centres published in 2005 found that general anaesthesia was used in only forty-seven percent of children over six months of age and twenty-two percent of younger infants, and that many infants had no sedation at all.

Most anaesthesia for endoscopy now uses propofol. The Pediatric Sedation Research Consortium, reported by Biber and colleagues in 2015, described twelve thousand and thirty endoscopies under sedation, ninety-seven percent of them with propofol. Adverse events occurred in about five percent. The most common were a drop in oxygen saturation in one and a half percent, airway obstruction in one percent, coughing in about one percent, and laryngospasm in about six in a thousand. There were no deaths and no cardiopulmonary resuscitations. The independent risk factors were age five years or younger, an American Society of Anesthesiologists physical status class of two or higher, an upper rather than lower endoscopy, obesity, and lower airway disease such as asthma. So the upper endoscopy in a small, obese or wheezy child is the case to plan the airway for.

Should the airway be secured with an endotracheal tube? Most procedures are done with a natural airway. In a Boston series of two hundred and fifty-eight children published by Hartjes and colleagues in 2021, about one in five was intubated, and upper endoscopy raised the odds of intubation more than four-fold, because the endoscope and the anaesthetist share the airway. Sedation adverse events occurred in three and a half percent and included three episodes of laryngospasm. A Texas comparison found fewer adverse events with propofol given by non-anaesthesiologists than with general anaesthesia, but the patients were very different, so this mostly shows how selection drives such comparisons. The practical lesson is to decide the airway plan with the anaesthetist before the list begins, and to secure it when the risk of aspiration is high, for example in bleeding or with a retained foreign body.

Fasting rules differ between Europe and the United States. The 2022 European paediatric fasting guideline from the European Society of Anaesthesiology and Intensive Care, ESAIC, led by Frykholm, allows clear fluids until one hour before anaesthesia, breast milk until three hours, formula or a light meal until four hours, and solid food until six hours. This is easy to remember as six, four, three, one. The 2017 guideline of the American Society of Anesthesiologists, ASA, asks

for two hours for clear fluids, four for breast milk, and six for formula and solids. Long fasts make children thirsty, hungry and sometimes hypoglycaemic without reducing aspiration, which is rare in elective cases. For endoscopy there is one more reason to allow clear fluids: a child who has just completed bowel preparation is already dehydrated.

Part four. Bowel preparation.

A poor bowel preparation is the most common reason a paediatric colonoscopy fails. In reviews by Hunter and Mamula, up to one in three children were inadequately prepared. In a multicentre observational study by Berger and colleagues in 2016, covering seven hundred and sixty-eight colonoscopies in twenty-eight centres, about seventy-three percent of preparations were good or excellent, so about one in four was not. A poor preparation hides small aphthous ulcers in Crohn disease, makes it harder to reach the terminal ileum, and often means a second anaesthetic. The PEnQuIN standards set a minimum target of at least eighty percent adequate preparations, at least ninety percent caecal intubation, and at least eighty-five percent terminal ileal intubation.

The ESPGHAN and ESGE guideline makes two strong recommendations based on high-quality evidence. It recommends a low-volume preparation, either polyethylene glycol with ascorbate or sodium picosulphate with magnesium citrate, and it recommends against sodium phosphate, which can cause serious electrolyte disturbances and kidney injury.

In North America, the most widely used agent is polyethylene glycol three three five zero without electrolytes, the same powder used daily for constipation, given at a higher dose. The 2014 NASPGHAN report by Pall and colleagues described several regimens, including a one-day regimen of about four grams per kilogram with a stimulant such as bisacodyl, a fixed dose of two hundred and thirty-eight grams in a sports drink for larger children, two-day regimens, and nasogastric polyethylene glycol with electrolytes for children who cannot drink the volume. The report also noted honestly that few paediatric studies had compared these regimens.

The evidence for polyethylene glycol three three five zero looks like this. Pashankar and colleagues in 2004 gave one and a half grams per kilogram per day for four days to forty-six children, and the preparation was good or excellent in ninety-one percent for the right colon and ninety-five percent for the left colon; nausea, abdominal pain and vomiting each affected about one child in nine. Safder and colleagues in 2008 used the same four-day regimen in one hundred and sixty-seven children, and about eighty-nine percent of those who completed it were adequately prepared; at least five stools a day and liquid stools in the last two days predicted success. Elitsur and colleagues in 2013 randomised children to the four-day regimen or a two-day regimen of two grams per kilogram per day plus bisacodyl, and the two-day regimen was at least as good, with adequate preparation in about seventy-four percent compared with fifty-eight percent, a difference that was not statistically significant. One-day regimens have also worked well: a retrospective series by Adamiak and colleagues reported effective preparation in ninety-three percent of two hundred and seventy-two children. In adolescents, Abbas and colleagues gave two hundred and thirty-eight grams in almost two litres of a sports drink; the preparation was effective in

seventy-seven percent and the terminal ileum was reached in eighty-four percent, but sixty percent had nausea or vomiting. Sahn and colleagues checked blood chemistry before and after a one-day regimen of four grams per kilogram in one hundred and fifty-five children; about a quarter developed mild hypokalaemia, none below three point three millimoles per litre, and a few had low blood glucose. So shorter regimens work as well as longer ones, but in small or unwell children on a one-day regimen it is sensible to check potassium and glucose.

Comparative trials show that the agents are similarly effective and differ mainly in how easy they are to take. In a randomised trial of two hundred and ninety-nine children aged two to eighteen published by Di Nardo and colleagues in 2014, polyethylene glycol with simethicone, polyethylene glycol with ascorbate, and sodium picosulphate with magnesium citrate were all non-inferior to standard high-volume polyethylene glycol with electrolytes, and picosulphate had the best tolerability and compliance. A systematic review by Gordon and colleagues in 2017 found no difference in efficacy between regimens, but children given high-volume polyethylene glycol with electrolytes needed a nasogastric tube far more often: forty-five of one hundred and seventeen, compared with two of one hundred and twenty-one on picosulphate. In a phase three trial published by Cuffari and colleagues in 2020, the full dose of sodium picosulphate with magnesium citrate worked in eighty-eight percent of children aged nine to twelve, compared with eighty-one percent for polyethylene glycol, while a half dose worked in only half. Children were about twice as likely to call picosulphate easy to drink. The product is approved in the United States from nine years of age. The lesson is to choose the regimen the child will actually finish, and then give it at full dose.

Part five. Choosing the instrument by weight.

The ESPGHAN and ESGE guideline states that the type of gastroscope or colonoscope should depend on the child's weight and age, and it uses ten kilograms or one year of age as the main dividing line. Below ten kilograms, upper endoscopy should use a gastroscope six millimetres or less in diameter, although a standard adult gastroscope may be considered if therapy such as variceal banding is needed. From ten kilograms, a standard adult gastroscope can be used, or a therapeutic gastroscope if needed. For colonoscopy below ten kilograms, a small gastroscope, an adult gastroscope or a paediatric colonoscope can be used; above ten kilograms, a paediatric or adult colonoscope. For ERCP, a paediatric duodenoscope of seven and a half millimetres is recommended below ten kilograms, and a therapeutic duodenoscope above it. For endoscopic ultrasound, small children need a miniprobe or a bronchial ultrasound scope, and an adult echoendoscope can be used from about fifteen kilograms or three years of age.

The 2012 report of the technology committee of the American Society for Gastrointestinal Endoscopy, ASGE, is similar but sets the colonoscope threshold a little higher, at twelve to fifteen kilograms. It notes that ultrathin paediatric gastroscopes, between about five and six millimetres in diameter with a two millimetre channel, limit suction, biopsy size and therapy, and that therapeutic gastroscopes, which are eleven to thirteen millimetres wide, should be avoided below ten to fifteen kilograms.

Part six. What the guidelines say.

When we compare North American and European documents, most points are aligned. Both prefer anaesthesia-supported endoscopy. Both avoid sodium phosphate in children. Both ask for upper endoscopy and ileocolonoscopy in every child with suspected inflammatory bowel disease, which in Europe is written into the revised Porto criteria of 2014. The Helicobacter guideline is a joint document, so the advice against test-and-treat is the same on both sides of the Atlantic.

There are some differences. North America relies more on polyethylene glycol three three five zero regimens, while Europe recommends low-volume polyethylene glycol with ascorbate or picosulphate. The American fasting guideline allows clear fluids until two hours before anaesthesia, while the European guideline allows them until one hour.

The material difference is coeliac disease. In Europe, a child with a TGA at least ten times normal and a positive EMA on a second sample can be diagnosed without endoscopy. The 2016 NASPGHAN clinical report on gluten-related disorders, by Hill and colleagues, still advised biopsy confirmation in most children. So the same child may have an endoscopy in one system and not in the other.

Part seven. What to do differently on Monday.

Let us return to the four children on the list. The nine-year-old with periumbilical pain, normal growth, normal blood tests and normal calprotectin has functional abdominal pain; she does not need an endoscopy, and her family needs an explanation and a plan. The six-year-old with a TGA fourteen times normal and a positive EMA on a second sample can be diagnosed with coeliac disease without biopsy under the European guideline. The fourteen-year-old with bloody diarrhoea and a high calprotectin needs both upper endoscopy and ileocolonoscopy, under general anaesthesia, with a low-volume preparation that he will actually finish. The seven-kilogram infant with haematemesis needs a gastroscope of six millimetres or less, under general anaesthesia with a secured airway, within twenty-four hours.

Here are the habits to take away. Ask what tissue or therapy the endoscopy will provide before you book it. Take systematic biopsies from normal-looking mucosa. Do not scope uncomplicated reflux, functional pain or constipation, and do not test for Helicobacter outside endoscopy. Use calprotectin to decide who needs a colonoscopy. Plan the airway with the anaesthetist for young, obese or wheezy children and for upper procedures. Let children drink clear fluids until the European limit of one hour, especially after bowel preparation. Choose a bowel preparation the child will finish, give it at full dose, check electrolytes when the child is small or unwell, and audit your own rate of adequate preparation against the target of eighty percent. And choose the endoscope by weight, with ten kilograms as the line to remember.

Questions to test yourself.

Question one. In the Colorado study of one thousand first upper endoscopies, how much histological disease would have been missed by biopsying only abnormal-looking mucosa?

Answer: about half, forty-eight and a half percent.

Question two. What are the ESPGHAN criteria for diagnosing coeliac disease without biopsy?

Answer: a TGA IgA of at least ten times the upper limit of normal, with a positive endomysial antibody in a second blood sample. In the ProCeDE study, the positive predictive value was ninety-nine point seven five percent.

Question three. What is the recommended diagnostic approach if *Helicobacter pylori* infection is suspected in a child? Answer: endoscopy with at least six biopsies, for histology and culture or PCR to guide therapy; non-invasive test-and-treat is not recommended.

Question four. What were the sensitivity and specificity of faecal calprotectin for inflammatory bowel disease in children in the 2010 meta-analysis? Answer: ninety-two and seventy-six percent.

Question five. What does the ESPGHAN and ESGE guideline say about sedation for diagnostic endoscopy? Answer: general anaesthesia, or deep sedation in a carefully monitored setting only if general anaesthesia is not available.

Question six. Name the independent risk factors for sedation adverse events in the Pediatric Sedation Research Consortium. Answer: age five years or younger, physical status class two or higher, upper endoscopy, obesity and lower airway disease.

Question seven. What are the European fasting times for children? Answer: six hours for solids, four for formula or a light meal, three for breast milk and one for clear fluids.

Question eight. What is the PEnQuIN minimum target for adequate bowel preparation, and which agent is the easiest for children to drink? Answer: at least eighty percent; sodium picosulphate with magnesium citrate.

Question nine. Below what weight should a gastroscope of six millimetres or less be used? Answer: below ten kilograms or under one year of age.

End of chapter.